

YIFAN ZHU

Northwestern Polytechnical University | Software Engineering Undergraduate,
Junior Year

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Laurent-Zhu

Xi'an, Shaanxi



EDUCATION

B.Eng. in Software Engineering

Northwestern Polytechnical University

2023.09 – Present

- **Major Courses:** Fundamentals of Programming, Advanced Mathematics, Data Structures, Software Engineering, Fundamental Software Development Training, Computer Networks, Fundamentals of Database Systems, Computer Operating Systems, Principles of Computer Organization, Deep Learning, Software Testing, Software Project Organization and Management, Algorithm Analysis and Design, Machine Learning
- **Academic Performance:** GPA 3.813/4.1, ranked 11/270 in the cohort after the first five semesters; ranked 12/270 in comprehensive evaluation
- **English Proficiency:** CET-4 613; CET-6 587; CET-6 Speaking Test: **Excellent**
- Awarded the **National Scholarship for Undergraduate Students** for **two consecutive years** among only 3 students in the cohort

RESEARCH EXPERIENCE

1. BrepLLM: Enabling Large Language Models to Understand Boundary Representations

ECCV 2026 Accepted, Fifth Author

2. Unified Multimodal Agent Framework for CadQuery-based 3D-CAD Generation, Understanding and Editing (Ongoing)

Ongoing

- Participating in a research project on a multimodal agent framework for 3D-CAD generation, understanding, and editing, using executable Python CadQuery programs as the unified modeling representation. My work focuses on high-quality CadQuery data construction, unified SFT data format design, model fine-tuning, and CAD-kernel-based automated evaluation and repair.

INTERNSHIP EXPERIENCE

Shenzhen Kuakua Jingling Technology Co., Ltd.

Back-end Developer

2025.07 – 2025.08

- Implemented workflow orchestration for a digital employee agent, enabling automatic generation of enterprise materials
- Assisted in developing integration interfaces for a municipal big data bureau project, covering the full process from requirements analysis to deployment
- Assisted my mentor in conducting research on AI multi-turn dialogue QA and knowledge-base products, and wrote a detailed research report that provided key references for the company's technology selection and product planning

SELECTED HONORS AND AWARDS

International Level

- Honorable Mention, 2026 Mathematical Contest in Modeling (MCM) (2026.05)

National Level

- National Scholarship for Undergraduate Students (2024.12)
- National Scholarship for Undergraduate Students (2025.12)
- Second Prize, Special Track of the 19th "Challenge Cup" National College Students' Extracurricular Academic Science and Technology Works Competition (2024.11)

Provincial / Regional Level

- Third Prize, 16th National College Students Mathematics Competition, Non-Mathematics Category A (2024.12)
- Third Prize, Northwest Regional Contest, 18th National Undergraduate Software Innovation Competition – Software Design Innovation Contest (2025.04)
- Third Prize, Shaanxi Division, 2025 11th National College Students Statistical Modeling Competition (2025.05)
- Second Prize, Shaanxi Division, 2025 Contemporary Undergraduate Mathematical Contest in Modeling (2025.11)

PROJECT EXPERIENCE

Dual-modal UAV Urban Traffic Perception and Intelligent Analysis System

Undergraduate Innovation Training Project

- Developed a UAV-based urban traffic object detection and visualization system using visible-light and infrared dual-modal fusion. The system targets five types of urban scenarios, including parking lots, main roads, and residential areas, and detects five categories of objects such as cars, trucks, and buses. A customized YOLOv8-OBB fusion model was trained on the public DroneVehicle dataset.
- Responsible for designing and implementing a three-layer Edge-Cloud-Dashboard architecture. The edge node integrates a dual-modal RGB+infrared YOLOv8-OBB fusion inference engine, simulates UAV flight views in five urban scenarios, and reports detection results and video streams to the cloud in real time at 15 fps through WebSocket. The FastAPI cloud service supports multi-node access and data forwarding, while the React dashboard supports both real-time streaming monitoring and offline prerecorded data replay.
- GitHub Repository

AgriGuard –Smart Agricultural Pest Management Platform

Agile Development Practice, Software Engineering Capstone Course

- End-to-end digital pest control system for agricultural production, serving three user roles (farmer, manager, field worker); integrates pest image recognition, RAG-based treatment recommendation, task scheduling, work-log traceability, and 7-day risk forecasting into a fully browser-accessible business workflow
- Stack: Vue 3 + Element Plus + ECharts (frontend), Django 5 + DRF + JWT (backend), YOLO for object detection and FAISS + LLM for RAG knowledge retrieval, MySQL; delivered via Scrum across 3 sprints with a 7-person cross-functional team, shipping 298+ backend automated test cases
- Responsible for backend API development of the pest recognition module, YOLO model training and deployment, and selected frontend pages, spanning recognition upload, task scheduling, work-log query, and risk prediction modules
- GitHub Repository

SELECTED COURSE GRADES

Course Name	Grade
Artificial Intelligence	
Deep Learning	91
Introduction to Image Processing and Computer Vision	90
Unmanned Aerial Vehicle System Engineering and Intelligent Perception Technology	90
Mathematics and Statistics	
Calculus I (Part I)	99
Calculus I (Part II)	98
Linear Algebra I	92
Discrete Mathematics	90
Probability Theory and Mathematical Statistics I	90
Complex Variables and Integral Transforms	90
Mathematical Modeling	95
Algorithm Analysis and Design	86
Signals and Linear Systems I	96

FUTURE PLAN

- In the future, I hope to systematically conduct research on large language models during my postgraduate studies, with a focus on frontier topics such as model architecture, multimodal understanding, and AI agents. After graduation, I hope to join a leading company or high-level laboratory in the field and engage in research and practical implementation related to large language models.

SKILLS

Python

C++

Java

Machine Learning

NLP

LLM

AI Agent

Reinforcement Learning